

# Systematic Hyperparameter Sensitivity Analysis of Random Forest: A Full Factorial Study on Classification and Regression Benchmarks

Thapakorn Sinlapakorn, Pirapong Singsathid \*

**Abstract.** Random Forest predictive performance depends critically on hyperparameter choice, yet practitioners lack quantitative guidance on which parameters most influence outcomes. Existing sensitivity studies are restricted to classification tasks, rely on local one-at-a-time methods that cannot capture interaction effects, or employ surrogate models rather than exhaustive experimentation. This study applies a full factorial design of 1,024 configurations, each evaluated via 10-fold cross-validation across 12 classification and 8 regression benchmark datasets, yielding 204,800 total model evaluations. Sensitivity is quantified by eta-squared ( $\eta^2$ ) effect sizes with 95% percentile bootstrap intervals, Friedman non-parametric tests, and Wilcoxon signed-rank post-hoc comparisons under Shaffer’s static correction. `max_depth` is the dominant parameter across both task types, with a mean  $\eta^2$  of 0.389 for classification and 0.729 for regression, appearing as `None` in 17 of 20 best-performing configurations within the tested grid. Sensitivity profiles diverge systematically by task: `max_features` plays a proportionally larger role in classification ( $\eta^2 = 0.268$ ) than regression ( $\eta^2 = 0.119$ ), while `max_depth` dominance is 87% larger in regression than in classification. The 20 datasets fall into three sensitivity regimes named by their dominant parameter, and which regime applies is predicted by sample size (Spearman  $\rho = 0.834$  against  $\eta^2(\text{max\_depth})$ ). These findings provide evidence-based prioritization of tuning effort: `n_estimators` and `min_samples_split` can safely be fixed at default values, `max_depth` should be left unconstrained rather than swept, and `max_features` is where tuning effort returns most.

**Keywords:** Random Forest, hyperparameter sensitivity, full factorial design, eta-squared, Friedman test, Wilcoxon signed-rank test, benchmark evaluation, classification, regression

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## 1 Introduction

Random Forest (RF) [1] consistently ranks among the most competitive off-the-shelf learning algorithms, yet its predictive performance depends on the choice of hyperparameters that govern tree construction and ensemble aggregation [10]. The cost of a poor choice is not marginal: across the benchmark used here, the spread between the worst and best of the 1,024 configurations reaches 0.375 in  $R^2$  and 0.149 in weighted F1. This motivates systematic investigation of which parameters drive performance and by how much.

Prior sensitivity studies share three limitations. They rely on local one-at-a-time designs or on surrogates fitted to observational records rather than on controlled experiment [13, 14]; they report which parameters matter, or whether tuning pays at all, rather than what share of performance variance each parameter explains [17]; and none covers regression.

This study addresses these gaps through four contributions:

1. A full factorial evaluation of all 1,024 RF hyperparameter configurations on 20 benchmark datasets spanning both classification and regression tasks, giving model-free estimates of every main effect and pairwise interaction.
2. A data-driven classification of all 20 datasets into three sensitivity regimes by dominant hyperparameter: `max_depth`-dominant, `max_features`-dominant, and `min_samples_leaf`-dominant, providing a basis for conditional tuning decisions.
3. Identification of systematic divergence between classification and regression sensitivity profiles: `max_depth` dominance is substantially larger in regression, a contrast invisible to classification-only benchmarks.
4. Evidence-based tuning recommendations separating the two parameters that repay tuning, `max_depth` and `max_features`, from the two that can be left at their defaults across every dataset tested, `n_estimators` and `min_samples_split`.

Section 2 reviews related work on RF hyperparameter analysis and sensitivity analysis methods. Section 3 describes the experimental methodology. Section 4 presents the results. Section 5 interprets the findings and their practical implications. Section 6 concludes the paper.

## 2 Related Work

### 2.1 Random Forest: Formal Model and Hyperparameter Structure

Random Forest [1] constructs an ensemble of  $T$  decision trees  $\{h_1, \dots, h_T\}$ , each grown on a bootstrap sample  $\mathcal{B}_t$  drawn with replacement from training set  $\mathcal{D}$  of  $n$  instances. At every internal node, the optimal split is selected over a randomly drawn subset of  $m \leq p$  features, where  $p$  is the total number of input features. For a query

point  $\mathbf{x}$ , the ensemble aggregates individual tree predictions through majority voting (classification) or averaging (regression):

$$\hat{y} = \arg \max_c \sum_{t=1}^T \mathbf{1}[h_t(\mathbf{x}) = c] \quad (1)$$

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T h_t(\mathbf{x}) \quad (2)$$

For the regression case, the expected squared prediction error of the ensemble decomposes as [1]:

$$\mathbb{E}[(\hat{y} - y)^2] = \underbrace{\sigma_\varepsilon^2}_{\text{irreducible}} + \underbrace{\text{Bias}^2(\hat{y})}_{\substack{\text{max\_depth} \\ \text{min\_samples\_leaf}}} + \underbrace{\rho_h \sigma_h^2}_{\substack{\text{max\_features} \\ \text{max\_depth}}} + \underbrace{\frac{1 - \rho_h}{T} \sigma_h^2}_{\text{n\_estimators}} \quad (3)$$

where  $\sigma_\varepsilon^2$  is the noise variance of the target, which no choice of hyperparameters can reduce,  $\sigma_h^2 = \text{Var}(h_t(\mathbf{x}))$  is the variance of any single tree prediction, and  $\rho_h = \text{Corr}(h_t(\mathbf{x}), h_{t'}(\mathbf{x}))$ ,  $t \neq t'$ , is the average pairwise inter-tree correlation. An analogous decomposition governs classification error [3].

None of the five hyperparameters appears in Eq. (3) directly. Each acts on one or more of its components, as the braces indicate, and the mechanism in each case is the following:

- **max\_depth** ( $D$ ) acts on two terms at once. A tree of depth  $D$  has at most  $2^D$  leaves and can therefore return at most  $2^D$  distinct values, which bounds how closely the ensemble can follow the response surface and sets  $\text{Bias}^2$ ; deeper trees also fit their bootstrap sample more closely, which raises  $\sigma_h^2$ . Because  $\sigma_h^2$  scales both variance terms, depth reaches the total error through both.
- **min\_samples\_leaf** and **min\_samples\_split** block splits that would leave too few observations in a child node. This holds  $\sigma_h^2$  down and raises  $\text{Bias}^2$ , the same pair of channels as depth. Their effect is largely subsumed by **max\_depth** when depth is constrained, and becomes the primary complexity control when it is not.
- **max\_features** ( $m$ ) acts on  $\rho_h$ . Drawing the candidate split features from a random subset of size  $m < p$  forces different trees to split on different variables, and that decorrelation is what lowers  $\rho_h$ . Restricting  $m$  from  $p$  to  $\lfloor \sqrt{p} \rfloor$  therefore reduces the correlated variance without directly inflating bias, so long as trees are deep enough to fit what the available features allow.
- **n\_estimators** ( $T$ ) enters only the uncorrelated variance term, as  $1/T$ . As  $T \rightarrow \infty$  that term vanishes, but the correlated floor  $\rho_h \sigma_h^2$  is untouched, so ensemble size yields strict diminishing returns.

The structure of Eq. (3) reveals an inherent asymmetry: beyond the irreducible  $\sigma_\varepsilon^2$ , only  $\rho_h$  and  $\sigma_h^2$  determine the error floor that persists as  $T \rightarrow \infty$ , while  $T$  itself provides a bounded benefit that saturates rapidly. This theoretical hierarchy motivates the empirical question at the center of this paper: how large are these effects in practice, and how do they vary across datasets and task types?

## 2.2 Prior Work and Research Gaps

Empirical investigations of RF hyperparameter importance have produced consistent but incomplete findings. Probst et al. [10] surveyed the influence of RF hyperparameters on predictive performance, identifying `max_features` and `min_samples_leaf` as commonly reported key parameters. A companion study by Probst et al. [11] quantified hyperparameter tunability across 38 algorithms and 38 OpenML datasets, finding that RF exhibits comparatively low global tunability relative to support vector machines, a statement about average behavior rather than about any particular parameter–dataset pair. van Rijn and Hutter [14] applied a functional ANOVA framework [6] to existing OpenML performance records, ranking `max_features` and `min_samples_leaf` as the dominant RF contributors. Mantovani et al. [7] covered 94 OpenML classification datasets, but for single-tree algorithms, which leaves the ensemble-specific parameters outside their scope. This line of work establishes which parameters matter on average; it does not measure how large each effect is.

Dedicated sensitivity analyses of RF hyperparameters have faced a methodological ceiling imposed by the tools they employed. Trithipkaiwanpon and Taetragool [13] applied one-at-a-time analysis and Latin Hypercube Sampling with ANOVA to four RF hyperparameters across four classification datasets, finding `min_samples_leaf` to be the most consistently sensitive parameter. OAT methods vary one parameter while holding the others at a chosen baseline, so they cannot detect interactions, and the effect they measure for any parameter depends on where the others were fixed [8]. The surrogate approach of van Rijn and Hutter [14] escapes that limitation but acquires another: a surrogate fitted to records in which `max_depth = None` is the prevailing default sees little variation in depth, and underestimates it accordingly. Section 5.1 returns to this point against the present estimates.

Large-scale empirical comparisons have framed the hyperparameter problem differently, asking whether tuning yields a worthwhile return rather than which parameters are responsible. Sipper [12] evaluated tuned versus default performance across 26 algorithms and 250 PMLB benchmark datasets using Bayesian optimization with 50 trials per run, finding that tuning provides modest average gains with substantial algorithm–dataset variation. Weerts et al. [17] introduced a tuning risk metric, the performance loss incurred when one hyperparameter is fixed at a default while the others are tuned, and showed across 59 OpenML-CC18 classification datasets that several can be fixed at computed defaults without measurable penalty. Both answer whether to tune; neither answers how much each parameter accounts for, and Weerts et al. leave interaction effects to future work.

The gap across these three lines of work is specific: no study combines a full

factorial experimental design with standardized effect-size quantification across both classification and regression benchmarks. The cost of that omission is measurable in the present results: main effects account for 0.83 of performance variance in classification and 0.93 in regression, leaving 0.17 and 0.07 to interactions and residual variation, which one- at-a-time designs and surrogate rankings cannot attribute. The present study fills this gap by exhaustively evaluating all 1,024 hyperparameter configurations on 12 classification and 8 regression benchmark datasets, and by quantifying sensitivity through  $\eta^2$  effect sizes, Friedman non-parametric tests, and Wilcoxon signed-rank post-hoc tests. The result is a complete, primary-evidence characterization of each hyperparameter’s role as predicted by Eq. (3).

### 3 Methodology

Equation (3) identifies  $\text{Bias}^2(\hat{y})$ ,  $\sigma_h^2$ , and  $\rho_h$  as the structural determinants of RF ensemble error, each modulated by a distinct subset of hyperparameters. The experiment below converts that characterisation into a measurable design.

#### 3.1 Experimental Design

Let  $\Theta = \{\theta_1, \dots, \theta_k\}$  denote the set of  $k$  hyperparameters under study, and let  $L_i = \{\ell_{i,1}, \dots, \ell_{i,q}\}$  be the finite set of  $q$  discrete levels assigned to parameter  $\theta_i$ . A full factorial design enumerates the complete Cartesian product

$$\mathcal{C} = L_1 \times L_2 \times \dots \times L_k, \quad |\mathcal{C}| = q^k \quad (4)$$

and evaluates every element of it independently. With  $k = 5$  and  $q = 4$  this yields  $4^5 = 1,024$  configurations per dataset. Because every combination of levels is observed exactly once, all  $k$  main effects and all  $\binom{k}{2} = 10$  pairwise interactions are estimable without assuming any functional form for the performance surface [8].

**Selection of the parameter set.** Four criteria determine which parameters enter  $\Theta$ .

1. **Mechanistic coverage.** Equation (3) contains three hyperparameter-controllable quantities: the ensemble-size term  $1/T$ , the inter-tree correlation  $\rho_h$ , and the single-tree variance  $\sigma_h^2$  together with  $\text{Bias}^2$ . The set `n_estimators`, `max_features`, `max_depth`, `min_samples_leaf` and `min_samples_split` places each of these under experimental control, leaving no term unmanipulated and no two parameters acting solely through the same channel.
2. **Comparability.** These are the parameters ranked by the prior studies [10, 14, 17, 13] against which the present results are placed, so the  $\eta^2$  shares reported below can be read directly against published importance rankings.

3. **Task-type invariance.** Each parameter carries an identical meaning and an identical admissible range in `RandomForestClassifier` and `RandomForestRegressor`, which is a precondition for the classification–regression comparison that motivates this study.
4. **Factorial tractability.** A complete design costs  $q^k$  evaluations per dataset, so every additional four-level parameter quadruples the experiment. At  $k = 5$  the design requires 204,800 model fits; at  $k = 6$  it would require 819,200, beyond what exhaustive enumeration permits while retaining 20 datasets and 10-fold cross-validation.

**Parameters deliberately excluded.** Every other performance-relevant argument of the estimator is held at its scikit-learn default, each for a reason traceable to the criteria above. `bootstrap` is a structural switch rather than a tuning knob: disabling it removes the resampling mechanism that defines a Random Forest, so the resulting model is a different estimator rather than a differently tuned one. `max_samples`, `ccp_alpha` and `max_leaf_nodes` violate mechanistic coverage by redundancy. The first modulates  $\rho_h$ , the channel already controlled by `max_features`; the other two act on the  $\sigma_h^2$ –Bias<sup>2</sup> channel already governed by `max_depth` and `min_samples_leaf`. Admitting them would partition one mechanism across more parameters rather than introduce a new one, deflating the measured  $\eta^2$  of depth without altering the underlying effect. The split `criterion` violates task-type invariance: its levels are not comparable across the two settings (`gini`, `entropy` and `log_loss` for classification; `squared_error`, `absolute_error`, `friedman_mse` and `poisson` for regression), so it cannot enter a single factorial design shared by both. `class_weight` violates the same criterion, applying to classification alone, and modifies the training objective rather than the ensemble structure. What these exclusions imply for the interpretation of the reported effect sizes is stated in Section 5.4.

Table 1 lists the five hyperparameters, their four levels and the scikit-learn defaults. Two levels resolve only a linear contrast and three add a single curvature degree of freedom; neither can distinguish a saturating response from a monotone one, and saturation is precisely what Eq. (3) predicts for `n_estimators`. Four levels are the smallest grid on which saturation, non-monotonicity and an interior optimum are all observable, and the largest that keeps  $q^k$  within reach. Within each parameter the levels bracket the scikit-learn default and extend to the practical limit in the regularizing direction, so that the default is never an endpoint of the tested range. `max_features = 1.0` is included deliberately as the boundary case in which every feature is available at every split and  $\rho_h$  attains its maximum, fixing the upper endpoint of the correlation channel; `max_depth = None` plays the corresponding role for unconstrained tree growth.

All experiments use scikit-learn 1.7.2 [9] under Python 3.13, with `random_state = 42` and `n_jobs = -1` throughout; the statistical procedures of Section 3.3 use SciPy 1.16.3 [16].

**Table 1:** Hyperparameter grid: five parameters at four levels each, yielding 1,024 configurations.

| Hyperparameter                 | Description            | Levels               | Default |
|--------------------------------|------------------------|----------------------|---------|
| <code>n_estimators</code>      | Ensemble size $T$      | 50, 100, 200, 500    | 100     |
| <code>max_depth</code>         | Maximum tree depth $D$ | None, 5, 10, 20      | None    |
| <code>max_features</code>      | Features per split $m$ | sqrt, log2, 0.5, 1.0 | sqrt    |
| <code>min_samples_split</code> | Min. samples to split  | 2, 5, 10, 20         | 2       |
| <code>min_samples_leaf</code>  | Min. samples at leaf   | 1, 2, 5, 10          | 1       |

### 3.2 Benchmark Datasets

Twenty benchmark datasets are used, 12 classification and 8 regression problems drawn from the scikit-learn built-in collection and from OpenML [15]. They were selected to vary the properties that plausibly moderate hyperparameter sensitivity rather than to maximize their number: sample size spans more than two orders of magnitude ( $n = 150$  to  $n = 21,613$ ), dimensionality ranges from  $p = 4$  to  $p = 64$ , and the collection includes balanced and imbalanced targets, binary and many-class problems, and purely numeric as well as mixed numeric–categorical feature sets. All are publicly distributed, so the benchmark is reproducible without private data, and it overlaps the collections used by the studies against which the findings are compared in Section 5. The 12/8 split between task types was not imposed by design; because  $\eta^2$ , the Friedman test and the post-hoc comparisons are each computed within a dataset, unequal group sizes affect the precision of the cross-dataset means reported in Section 4.1 but not the validity of the per-dataset estimates.

Tables 2 and 3 summarize the two groups. Categorical features in `credit-g` and `abalone` were encoded ordinally prior to model training, and a stratified random sample of 10,000 instances was drawn from `covertype` to keep computation tractable.

### 3.3 Evaluation and Statistical Analysis

**Cross-validation protocol.** Let  $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$  denote a dataset of  $n$  instances. Each configuration  $\mathbf{c} \in \mathcal{C}$  is evaluated by 10-fold cross-validation:  $\mathcal{D}$  is partitioned into folds  $\{F_1, \dots, F_{10}\}$  of approximately equal size, and the model is trained on  $\mathcal{D} \setminus F_j$  and evaluated on  $F_j$  for each  $j$ . Classification folds are stratified, so that each preserves the class proportions of the full dataset, i.e.  $|F_j \cap \mathcal{D}_c|/|F_j| \approx n_c/n$  for every class  $c$ . A single partition was preferred to repeated cross-validation because the available model fits were allocated to breadth:  $r$  repetitions would multiply the cost by  $r$ , whereas the same number of fits buys complete enumeration of the configuration space on all 20 datasets. The cost of that choice is taken up in Section 5.4.

**Performance metrics.** For each configuration  $\mathbf{c} \in \mathcal{C}$  and fold  $j$ , a scalar performance score  $M_j(\mathbf{c})$  is recorded. For classification tasks, the primary metric is the

**Table 2:** Classification benchmark datasets (12 datasets).

| Dataset       | OpenML<br>ID | Samples<br>$n$      | Features<br>$p$ | Classes | Characteristics                      |
|---------------|--------------|---------------------|-----------------|---------|--------------------------------------|
| Iris          | —            | 150                 | 4               | 3       | Small, balanced                      |
| Wine          | —            | 178                 | 13              | 3       | Small, balanced                      |
| Ionosphere    | 59           | 351                 | 34              | 2       | Binary, noisy features               |
| Vehicle       | 54           | 846                 | 18              | 4       | Multi-class                          |
| Breast Cancer | —            | 569                 | 30              | 2       | Binary, medical                      |
| Credit-G      | 31           | 1,000               | 20              | 2       | Imbalanced, categorical              |
| Banknote      | 1462         | 1,372               | 4               | 2       | Low-dimensional binary               |
| Digits        | —            | 1,797               | 64              | 10      | High-dimensional, multi-class        |
| Segment       | 36           | 2,310               | 19              | 7       | Many classes                         |
| Phoneme       | 1489         | 5,404               | 5               | 2       | Medium-scale binary                  |
| Covertypes    | 180          | 10,000 <sup>†</sup> | 54              | 7       | Large-scale, imbalanced              |
| Pol           | 722          | 15,000              | 26              | 2       | High-dimensional binary <sup>‡</sup> |

— Source: sklearn built-in. <sup>†</sup>Stratified sample from 581,012 instances (random\_state = 42). <sup>‡</sup>OpenML 722 is a binarized version of a regression dataset: the numeric target is split into two classes at its mean.

**Table 3:** Regression benchmark datasets (8 datasets).

| Dataset            | OpenML<br>ID | Samples<br>$n$ | Features<br>$p$ | Characteristics      |
|--------------------|--------------|----------------|-----------------|----------------------|
| Diabetes           | —            | 442            | 10              | Small-scale, medical |
| Abalone            | 183          | 4,177          | 8               | Categorical feature  |
| Kin8nm             | 189          | 8,192          | 8               | Nonlinear dynamics   |
| Elevators          | 216          | 16,599         | 18              | Large-scale          |
| Bike Sharing       | 42712        | 17,389         | 12              | Temporal pattern     |
| Houses             | 537          | 20,640         | 8               | Real estate pricing  |
| California Housing | —            | 20,640         | 8               | Geographic features  |
| House Sales        | 42165        | 21,613         | 17              | Mixed feature types  |

— Source: sklearn built-in.



weighted F1-score:

$$\text{F1}_w = \sum_{c=1}^C \frac{n_c}{n} \cdot \frac{2P_c R_c}{P_c + R_c} \quad (5)$$

where  $P_c$  and  $R_c$  are the per-class precision and recall,  $n_c$  is the number of true instances of class  $c$ , and  $n = \sum_c n_c$ . Weighted averaging accounts for class imbalance without discarding information from minority classes. For regression tasks, the primary metric is the coefficient of determination:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (6)$$

which is scale-free and therefore comparable across datasets with different target ranges. Accuracy and RMSE were also recorded for every configuration and are released with the per-dataset result files; they are not analysed here, because  $\eta^2$  computed on them reproduces the ranking obtained from the primary metrics. For each configuration the mean  $\bar{M}(\mathbf{c}) = \frac{1}{10} \sum_{j=1}^{10} M_j(\mathbf{c})$  and standard deviation  $s_M(\mathbf{c})$  across folds are retained.

**Eta-squared effect size.** The sensitivity of hyperparameter  $\theta_i$  on dataset  $\mathcal{D}$  is quantified by the eta-squared ( $\eta^2$ ) effect size, an ANOVA-style measure of the proportion of total performance variance attributable to variation in  $\theta_i$ :

$$\eta^2(\theta_i) = \frac{SS_{\text{between}}(\theta_i)}{SS_{\text{total}}} = \frac{\sum_{\ell \in L_i} n_\ell (\bar{M}_\ell - \bar{M})^2}{\sum_{\mathbf{c} \in \mathcal{C}} (\bar{M}(\mathbf{c}) - \bar{M})^2} \quad (7)$$

where  $\bar{M}_\ell$  is the mean performance over all configurations at level  $\ell$  of  $\theta_i$ ,  $n_\ell = |\mathcal{C}|/q$  is the number of such configurations, and  $\bar{M}$  is the grand mean over all  $|\mathcal{C}|$  configurations. The denominator  $SS_{\text{total}}$  includes variance contributions from all five hyperparameters, their interactions and residual variation, which makes  $\eta^2$  a marginal measure: it answers what fraction of observable performance variance is attributable to  $\theta_i$  alone. Being marginal, the five main effects need not sum to unity; over dataset means they account for 0.83 of total variance in classification and 0.93 in regression, the remainder being interaction and residual variance.  $\eta^2$  is preferred here to the functional-ANOVA importance of van Rijn and Hutter [14] for a reason specific to the design: because every cell of the factorial is observed, the sum-of-squares decomposition in Eq. (7) is exact and needs no surrogate model to interpolate unobserved configurations, and it reports the result on a bounded  $[0, 1]$  scale comparable across datasets and across the two performance metrics used here.

**Uncertainty of the cross-dataset means.** Each parameter’s headline  $\eta^2$  is a mean over datasets, so the relevant sampling unit for its precision is the dataset. A 95% percentile bootstrap interval is obtained by resampling the per-dataset  $\eta^2$  values

with replacement within each task type over  $B = 10,000$  resamples at seed 42, and taking the 2.5th and 97.5th percentiles of the resulting distribution of means. The interval quantifies sensitivity to the particular benchmark collection assembled here; it does not propagate uncertainty from the single cross-validation partition on which each  $\eta^2$  is computed, which is a separate limitation discussed in Section 5.4.

**Friedman test.** For each hyperparameter  $\theta_i$  and dataset  $\mathcal{D}$ , the Friedman test [4] assesses whether the  $q = 4$  level-specific performance distributions differ significantly; this rank-based protocol, followed by pairwise post-hoc comparison, is the standard for comparing methods over multiple datasets [2]. Configurations are grouped by level and ranked within each group, and the test statistic

$$\chi_F^2 = \frac{12}{N_b q(q+1)} \sum_{j=1}^q R_j^2 - 3N_b(q+1) \quad (8)$$

follows a  $\chi^2$  distribution with  $q - 1$  degrees of freedom under the null hypothesis of equal distributions, where  $R_j$  is the sum of ranks for level  $j$  and  $N_b = |\mathcal{C}|/q$  is the number of configuration blocks. Significance is assessed at  $\alpha = 0.05$ .

**Wilcoxon signed-rank post-hoc test.** Where the Friedman test rejects the null hypothesis, all  $\binom{q}{2} = 6$  pairwise level comparisons are conducted using the Wilcoxon signed-rank test [18], identifying the specific level contrasts that drive the observed effect. The six comparisons within a hyperparameter form one family, and family-wise error is controlled by Shaffer’s static procedure [5], which exploits the logical relations among pairwise hypotheses: with  $q = 4$  levels the number of simultaneously true null hypotheses can only be 0, 1, 2, 3 or 6, these being the pair counts of the five partitions of four levels into equality classes. The ordered  $p$ -values are therefore compared step-down against  $\alpha/6, \alpha/3, \alpha/3, \alpha/3, \alpha/2, \alpha$  rather than against the uniformly conservative  $\alpha/6$  of a Bonferroni or Nemenyi correction. Table 9 reports both the uncorrected and the Shaffer-adjusted counts. The Bergmann–Hommel dynamic procedure is uniformly at least as powerful as Shaffer’s static procedure [5], so the adjusted counts are a lower bound on what that procedure would return.

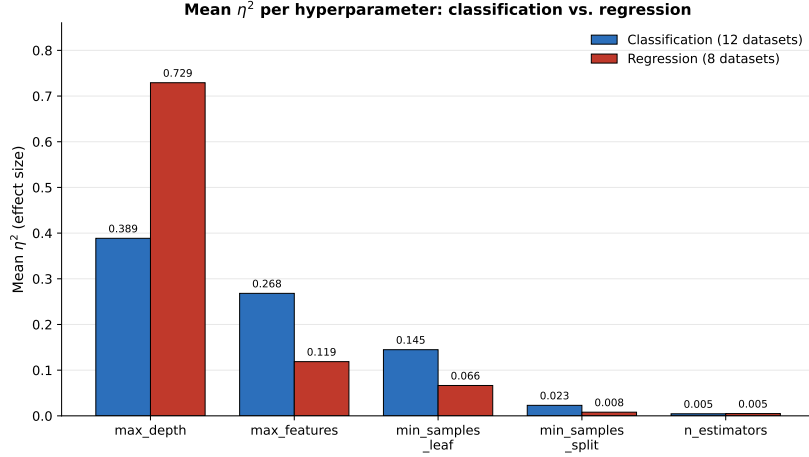
## 4 Results

### 4.1 Global Sensitivity Profiles

Across all 20 benchmark datasets, `max_depth` is the dominant hyperparameter in both task types, `max_features` ranks second, and `min_samples_leaf` ranks third; `min_samples_split` and `n_estimators` are consistently negligible. This ranking is stable across task types but with markedly different magnitudes, as Figures 1 and 2 show. Bootstrap intervals on these means (Tables 4 and 6) are wide, as 12 and 8 datasets allow, and in classification the intervals for `max_depth` and `max_features` overlap. The

separation that matters for the tuning recommendation survives nonetheless: neither leading interval reaches the classification upper bound for `min_samples_split`, 0.042, or for `n_estimators`, 0.008, and in regression `max_depth` is disjoint from every other parameter. Even the most favourable resampling of this benchmark leaves the `n_estimators` interval disjoint from those of both leading parameters, in both task types.

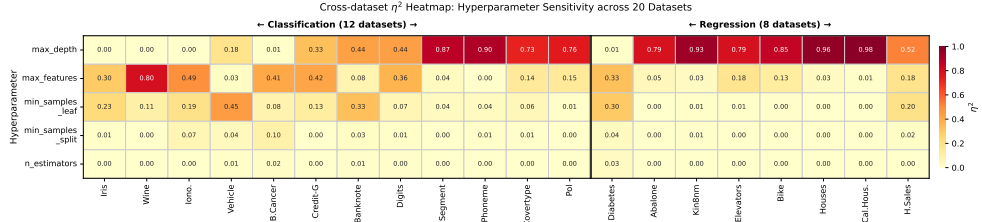
Pol warrants a robustness check, because it is not a native classification problem: OpenML 722 binarizes a numeric target at its mean, and its profile is correspondingly regression-like, carrying the third highest  $\eta^2(\text{max\_depth})$  of any classification dataset. Excluding it lowers the classification mean for `max_depth` from 0.389 to 0.355 and shifts the other four means by less than 0.012, leaving the rank order unchanged. The task-type divergence of Section 4.3 does not depend on it either, and in fact widens without it, from 87% to 105%. Pol is retained in the reported figures, and every classification mean in this paper should be read as including one binarized regression problem.



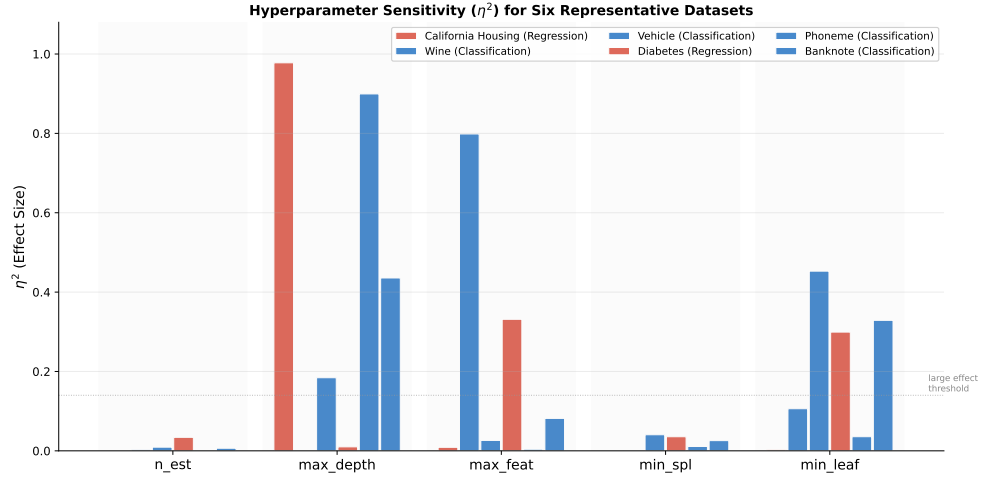
**Figure 1:** Mean  $\eta^2$  per hyperparameter, classification (blue) against regression (red), averaged over the 12 and 8 datasets of each task type.

Classifying each dataset by its dominant hyperparameter yields three regimes. Thirteen datasets are **max\_depth-dominant**: every regression dataset except Diabetes, together with Banknote, Digits, Segment, Phoneme, Covertypes and Pol. Six are **max\_features-dominant**: Iris, Wine, Ionosphere, Breast Cancer, Credit-G and Diabetes. Vehicle alone is **min\_samples\_leaf-dominant**. Figure 3 illustrates a representative example of each regime.

The Friedman test rejects the null at  $\alpha = 0.05$  for 97 of the 100 hyperparameter-dataset pairs. Each of the three remaining cases admits a dataset-specific explanation, given in Section 4.2: `n_estimators` on Ionosphere ( $\chi_F^2 = 5.21$ ,  $p = 0.163$ ), `max_features` on Phoneme ( $\chi_F^2 = 0.43$ ,  $p = 0.935$ ), and `max_depth` on Diabetes ( $\chi_F^2 = 6.29$ ,  $p = 0.108$ ).



**Figure 2:** Cross-dataset  $\eta^2$  heatmap over all 20 benchmark datasets. Rows are hyperparameters ordered by mean  $\eta^2$ ; columns are datasets, classification left and regression right. Darker red indicates higher sensitivity.



**Figure 3:** Hyperparameter sensitivity ( $\eta^2$ ) for six representative datasets: California Housing and Diabetes (regression, contrasting depth sensitivity), Wine and Phoneme (classification, contrasting feature sensitivity), and Vehicle and Banknote (classification, contrasting leaf against depth dominance). The dotted line marks a conventional large-effect threshold,  $\eta^2 = 0.14$ .

## 4.2 Main Effects and Level Contrasts

Effect sizes are reported in Tables 4 and 6, Friedman  $p$ -values in Tables 5 and 7, and Wilcoxon level contrasts in Tables 8 and 9.

**Table 4:** Eta-squared ( $\eta^2$ ) effect sizes for each hyperparameter across 12 classification datasets. Bold indicates the highest  $\eta^2$  per row.

| Dataset       | n_est       | max_d         | max_f         | mss         | msl           |
|---------------|-------------|---------------|---------------|-------------|---------------|
| Iris          | 0.0023      | 0.0000        | <b>0.2962</b> | 0.0081      | 0.2349        |
| Wine          | 0.0030      | 0.0000        | <b>0.7985</b> | 0.0010      | 0.1061        |
| Ionosphere    | 0.0021      | 0.0033        | <b>0.4915</b> | 0.0661      | 0.1930        |
| Vehicle       | 0.0092      | 0.1842        | 0.0262        | 0.0405      | <b>0.4528</b> |
| Breast Cancer | 0.0211      | 0.0091        | <b>0.4099</b> | 0.1009      | 0.0821        |
| Credit-G      | 0.0035      | 0.3287        | <b>0.4170</b> | 0.0036      | 0.1260        |
| Banknote      | 0.0065      | <b>0.4356</b> | 0.0817        | 0.0257      | 0.3288        |
| Digits        | 0.0047      | <b>0.4405</b> | 0.3649        | 0.0083      | 0.0683        |
| Segment       | 0.0003      | <b>0.8727</b> | 0.0368        | 0.0030      | 0.0385        |
| Phoneme       | 0.0015      | <b>0.8994</b> | 0.0040        | 0.0109      | 0.0356        |
| Covertypes    | 0.0001      | <b>0.7328</b> | 0.1435        | 0.0071      | 0.0579        |
| Pol           | 0.0000      | <b>0.7571</b> | 0.1473        | 0.0009      | 0.0137        |
| <b>Mean</b>   | 0.0045      | 0.3886        | 0.2681        | 0.0230      | 0.1448        |
| <b>95% CI</b> | 0.002–0.008 | 0.200–0.588   | 0.146–0.403   | 0.008–0.042 | 0.078–0.222   |
| <b>Rank</b>   | 5           | 1             | 2             | 4           | 3             |

n\_est: n\_estimators, max\_d: max\_depth, max\_f: max\_features,  
mss: min\_samples\_split, msl: min\_samples\_leaf.

**max\_depth.** `max_depth` leads in both task types, and the gain it offers is concentrated almost entirely in one step. Depth-5 configurations are significantly worse than every deeper setting on 18 of 20 datasets (Shaffer-corrected; 19 uncorrected, Table 9), a near-universal performance floor produced by the  $2^D$  leaf bound of Section 2.1. Contrasts among the deeper levels are far less consistent, 7 to 12 of 20, so the gain lies in escaping severe restriction rather than in choosing among deep settings; Figure 4 shows the corresponding steep rise from depth 5 to depth 10 and a plateau thereafter.

Sensitivity is highest on large classification datasets with complex decision boundaries, where Phoneme, Segment, Pol and Covertypes all exceed  $\eta^2 = 0.73$ , and it is near-universal in regression. Diabetes is the single exception ( $p = 0.108$ ): with 442 samples and a low-complexity response surface, a depth constraint has little left to bind.

**max\_features.** `max_features` ranks second in both task types, and the Wilcoxon contrasts give it a two-tier structure. 1.0 against `sqrt` is significant on 19 of 20 datasets, so full-feature trees are broadly inferior to the random subspace default; `log2` against `sqrt` is significant on only 9, so the two heuristics are largely inter-

**Table 5:** Friedman test  $p$ -values for each hyperparameter across 12 classification datasets. Asterisk (\*) denotes significance at  $\alpha = 0.05$ .

| Dataset       | n_est   | max_d   | max_f   | mss     | msl     |
|---------------|---------|---------|---------|---------|---------|
| Iris          | <0.001* | 0.007*  | <0.001* | <0.001* | <0.001* |
| Wine          | <0.001* | 0.001*  | <0.001* | <0.001* | <0.001* |
| Ionosphere    | 0.163   | <0.001* | <0.001* | <0.001* | <0.001* |
| Vehicle       | <0.001* | <0.001* | <0.001* | <0.001* | <0.001* |
| Breast Cancer | <0.001* | <0.001* | <0.001* | <0.001* | <0.001* |
| Credit-G      | <0.001* | <0.001* | <0.001* | <0.001* | <0.001* |
| Banknote      | <0.001* | <0.001* | <0.001* | <0.001* | <0.001* |
| Digits        | <0.001* | <0.001* | <0.001* | <0.001* | <0.001* |
| Segment       | 0.046*  | <0.001* | <0.001* | <0.001* | <0.001* |
| Phoneme       | <0.001* | <0.001* | 0.935   | <0.001* | <0.001* |
| Coverttype    | <0.001* | <0.001* | <0.001* | <0.001* | <0.001* |
| Pol           | <0.001* | <0.001* | <0.001* | <0.001* | <0.001* |

n\_est: n\_estimators, max\_d: max\_depth, max\_f: max\_features,  
mss: min\_samples\_split, msl: min\_samples\_leaf.

**Table 6:** Eta-squared ( $\eta^2$ ) effect sizes for each hyperparameter across 8 regression datasets. Bold indicates the highest  $\eta^2$  per row.

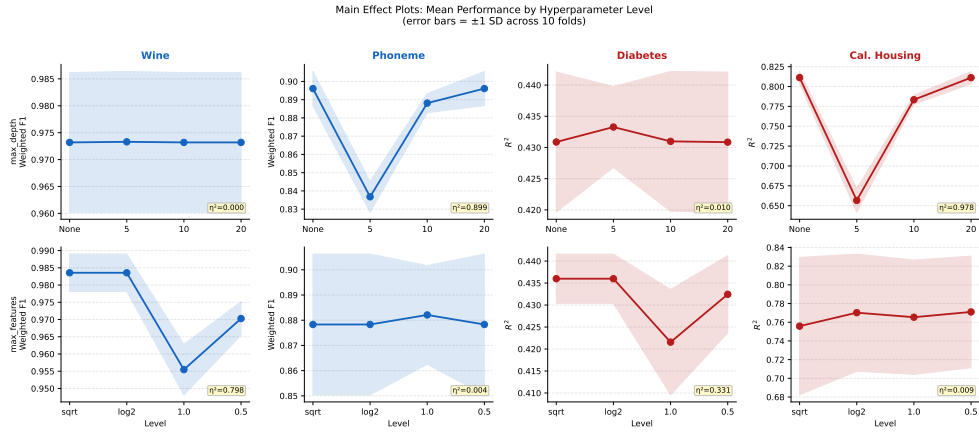
| Dataset            | n_est       | max_d         | max_f         | mss         | msl         |
|--------------------|-------------|---------------|---------------|-------------|-------------|
| Diabetes           | 0.0340      | 0.0100        | <b>0.3314</b> | 0.0355      | 0.2992      |
| Abalone            | 0.0034      | <b>0.7881</b> | 0.0533        | 0.0015      | 0.0039      |
| Kin8nm             | 0.0003      | <b>0.9340</b> | 0.0340        | 0.0063      | 0.0115      |
| Elevators          | 0.0001      | <b>0.7925</b> | 0.1796        | 0.0021      | 0.0123      |
| Bike Sharing       | 0.0002      | <b>0.8523</b> | 0.1274        | 0.0006      | 0.0046      |
| Houses             | 0.0002      | <b>0.9582</b> | 0.0314        | 0.0004      | 0.0022      |
| California Housing | 0.0002      | <b>0.9778</b> | 0.0088        | 0.0004      | 0.0029      |
| House Sales        | 0.0015      | <b>0.5205</b> | 0.1834        | 0.0175      | 0.1952      |
| <b>Mean</b>        | 0.0050      | 0.7292        | 0.1187        | 0.0080      | 0.0665      |
| <b>95% CI</b>      | 0.000–0.013 | 0.496–0.902   | 0.053–0.196   | 0.001–0.017 | 0.005–0.150 |
| <b>Rank</b>        | 5           | 1             | 2             | 4           | 3           |

n\_est: n\_estimators, max\_d: max\_depth, max\_f: max\_features,  
mss: min\_samples\_split, msl: min\_samples\_leaf.

**Table 7:** Friedman test  $p$ -values for each hyperparameter across 8 regression datasets. Asterisk (\*) denotes significance at  $\alpha = 0.05$ .

| Dataset            | n_est   | max_d   | max_f   | mss     | msl     |
|--------------------|---------|---------|---------|---------|---------|
| Diabetes           | <0.001* | 0.108   | <0.001* | <0.001* | <0.001* |
| Abalone            | <0.001* | <0.001* | <0.001* | <0.001* | <0.001* |
| Kin8nm             | <0.001* | <0.001* | <0.001* | <0.001* | <0.001* |
| Elevators          | <0.001* | <0.001* | <0.001* | <0.001* | <0.001* |
| Bike Sharing       | <0.001* | <0.001* | <0.001* | <0.001* | <0.001* |
| Houses             | <0.001* | <0.001* | <0.001* | <0.001* | <0.001* |
| California Housing | <0.001* | <0.001* | <0.001* | <0.001* | <0.001* |
| House Sales        | <0.001* | <0.001* | <0.001* | <0.001* | <0.001* |

n\_est: n\_estimators, max\_d: max\_depth, max\_f: max\_features,  
mss: min\_samples\_split, msl: min\_samples\_leaf.



**Figure 4:** Main effect plots for selected classification (blue) and regression (red) datasets. Each panel gives the mean primary metric as a function of hyperparameter level, with shaded bands at  $\pm 1$  standard deviation across the 10 folds; yellow badges give  $\eta^2$  for each parameter–dataset combination.

changeable. Random subspace selection, not the particular fraction chosen, is the decisive mechanism. Table 8 shows the pattern on Wine, where `sqrt` and `log2` tie on every fold while all three restricted levels outperform `max_features = 1.0` at  $p < 0.001$ .

The effect is largest on smaller, low-dimensional classification datasets where depth is not the primary bottleneck: Wine, Ionosphere, Credit-G and Breast Cancer all exceed  $\eta^2 = 0.40$ . The non-significant case, Phoneme ( $p = 0.935$ ), follows from its 5-feature space, where `sqrt` selects 2 features and `1.0` selects 5, a difference too small to affect ensemble diversity.

**min\_samples\_leaf.** `min_samples_leaf` ranks third, with its effect concentrated on small datasets where leaf-node overfitting is the primary failure mode: Vehicle, Banknote and Iris are the only classification datasets above  $\eta^2 = 0.20$ , and of the nine datasets with  $n > 5,000$  only House Sales exceeds  $\eta^2 = 0.06$ . Vehicle is the sole `min_samples_leaf`-dominant dataset in the benchmark; with roughly 210 instances per class, raising the minimum leaf size from 1 to 2 or 5 supplies regularization that a depth constraint cannot replicate. The Wilcoxon counts place the effect at large leaf sizes: 20 of 20 datasets for 1 against 10, but 15 of 20 for 1 against 2.

**min\_samples\_split and n\_estimators.** Both occupy ranks 4 and 5, with cross-dataset mean  $\eta^2$  below 0.025, yet Wilcoxon detects significant level differences on most datasets: 17 to 20 of 20 for the extreme `min_samples_split` contrasts and 13 to 17 for `n_estimators`. The differences are statistically real but practically negligible. Their main effect curves in Figure 4 are nearly flat throughout, and neither is a worthwhile tuning target under the conditions of this study.

### 4.3 Classification–Regression Divergence

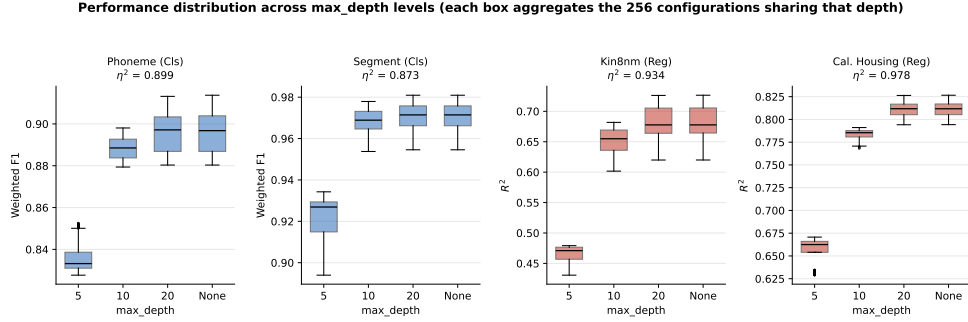
The most consequential structural finding is the systematic divergence between the two task types. `max_depth` dominance is 87% larger in regression than in classification, while `max_features` runs the other way, mattering more in classification. Section 5.1 takes up the mechanism. The divergence is invisible to studies restricted to classification benchmarks.

### 4.4 Pairwise Interactions

The dominant pairwise interaction is between `max_depth` and `min_samples_leaf`. Figure 5 documents the depth effect across levels; Figure 6 maps the interaction structure for contrasting dataset–regime pairs.

The distributions in Figure 5 separate the two regimes cleanly. At `max_depth = 5` the interquartile range barely overlaps those at deeper settings, and the best configurations at depth 5 do not reach the median of unconstrained trees: severe depth restriction acts as a hard performance ceiling rather than a soft regularizer, so no com-





**Figure 5:** Distribution of the primary metric across `max_depth` levels for four representative datasets; each box aggregates the 256 configurations sharing that depth value.

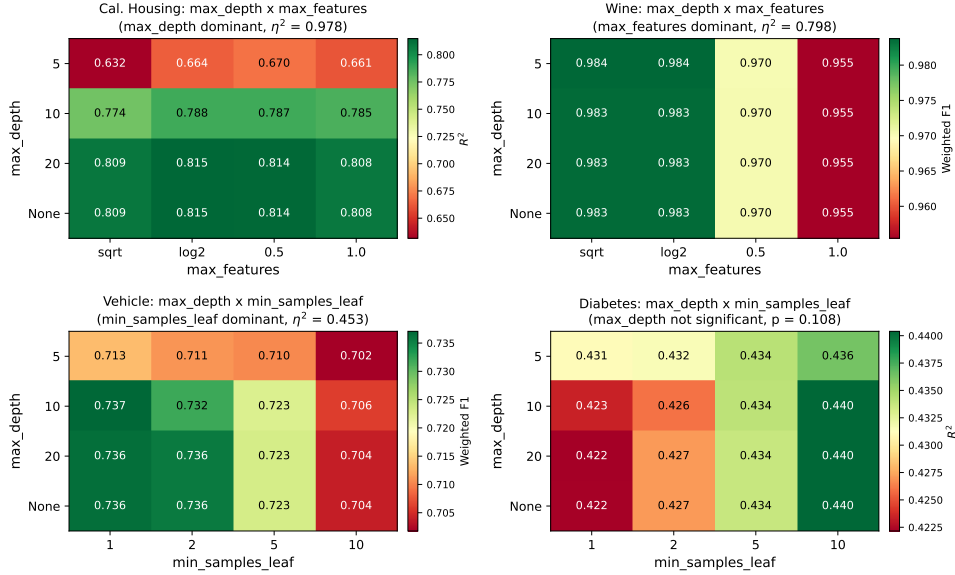
bination of the remaining four parameters can compensate. At depth 10 and 20 the interquartile ranges overlap `max_depth` = None substantially, placing the saturation point between depth 10 and 20 for these datasets.

Figure 6 shows that the interaction structure is regime-specific rather than universal. California Housing varies almost entirely along the depth axis, from 0.632 at depth 5 to 0.809 unconstrained, while Wine varies almost entirely along the feature axis and is flat in depth: the dominant parameter sets the direction of the gradient, and the second shifts performance uniformly rather than conditionally. Interaction is therefore weak wherever a single parameter dominates.

Interaction becomes substantive once depth stops binding. On Vehicle the spread along `min_samples_leaf` is 0.012 in the `max_depth` = 5 row and roughly 0.032 in each of the three deeper rows: a depth-5 tree is already too coarse for the leaf constraint to bite, and above that threshold leaf-level regularization becomes the primary remaining control on tree complexity. The change is a threshold at depth 5 rather than a gradual one, which is why `min_samples_leaf` attains its highest  $\eta^2$  on datasets whose best configuration leaves `max_depth` unconstrained. Diabetes shows the same leaf gradient without the depth one: performance rises from 0.422 to 0.440 across `min_samples_leaf` while the depth axis is flat, consistent with its non-significant Friedman result for `max_depth`.

The same conditional logic applies to `max_features` on high-dimensional classification data such as Digits and Breast Cancer, where restricted feature subsets benefit most from unconstrained depth: deep trees compensate for the reduced feature set at each split, and pairing `max_features` = 1.0 with shallow depth yields the weakest configurations.

Pairwise interaction heatmaps for four representative datasets



**Figure 6:** Pairwise interaction heatmaps for four representative datasets, each panel titled with the pair it shows. Top row: `max_depth`  $\times$  `max_features` on California Housing (`max_depth`-dominant) and Wine (`max_features`-dominant). Bottom row: `max_depth`  $\times$  `min_samples_leaf` on Vehicle (`min_samples_leaf`-dominant) and Diabetes (`max_depth` not significant). Cells give the mean primary metric over the 64 configurations sharing that level pair; the colour scale runs per panel from minimum (red) to maximum (green).

**Table 8:** Wilcoxon signed-rank post-hoc test results for `max_features` on the Wine dataset (primary metric: weighted F1-score). Significant differences ( $p < 0.05$ ) are indicated by \*.

| Level 1 | Level 2 | Statistic | $p$ -value       | Sig. |
|---------|---------|-----------|------------------|------|
| sqrt    | log2    | 0.0       | N/A <sup>†</sup> | —    |
| sqrt    | 0.5     | 136.0     | <0.001           | *    |
| sqrt    | 1.0     | 0.0       | <0.001           | *    |
| log2    | 0.5     | 136.0     | <0.001           | *    |
| log2    | 1.0     | 0.0       | <0.001           | *    |
| 0.5     | 1.0     | 0.0       | <0.001           | *    |

<sup>†</sup>All pairwise differences are zero (complete ties);  $p$ -value undefined.

**Table 9:** Number of datasets (out of 20) in which each pairwise level comparison achieves Wilcoxon significance ( $p < 0.05$ ). Raw counts are uncorrected; Shaffer counts apply Shaffer’s static procedure to the six comparisons within each hyperparameter at a family-wise  $\alpha = 0.05$  (Section 3.3). Nine of the 30 contrasts lose one dataset under the correction; the other 21 are unchanged.

| Hyperparameter    | Level pair   | Raw | Shaffer |
|-------------------|--------------|-----|---------|
| max_depth         | 5 vs 10      | 19  | 18      |
|                   | 5 vs 20      | 19  | 18      |
|                   | 5 vs None    | 19  | 18      |
|                   | 10 vs 20     | 13  | 12      |
|                   | 10 vs None   | 12  | 12      |
|                   | 20 vs None   | 7   | 7       |
| max_features      | 1.0 vs sqrt  | 19  | 19      |
|                   | 1.0 vs log2  | 18  | 18      |
|                   | 0.5 vs sqrt  | 17  | 17      |
|                   | 0.5 vs 1.0   | 16  | 15      |
|                   | 0.5 vs log2  | 16  | 16      |
|                   | log2 vs sqrt | 9   | 9       |
| min_samples_leaf  | 1 vs 10      | 20  | 20      |
|                   | 2 vs 10      | 20  | 20      |
|                   | 2 vs 5       | 20  | 19      |
|                   | 5 vs 10      | 20  | 20      |
|                   | 1 vs 5       | 19  | 19      |
|                   | 1 vs 2       | 15  | 15      |
| min_samples_split | 2 vs 20      | 20  | 20      |
|                   | 10 vs 20     | 20  | 20      |
|                   | 5 vs 20      | 20  | 20      |
|                   | 2 vs 10      | 19  | 19      |
|                   | 5 vs 10      | 19  | 19      |
|                   | 2 vs 5       | 17  | 17      |
| n_estimators      | 50 vs 100    | 17  | 16      |
|                   | 50 vs 200    | 17  | 17      |
|                   | 50 vs 500    | 17  | 17      |
|                   | 100 vs 500   | 15  | 14      |
|                   | 100 vs 200   | 14  | 13      |
|                   | 200 vs 500   | 13  | 13      |

## 4.5 Best Configurations Within the Tested Grid

Throughout this subsection, best denotes the highest-scoring configuration among the 1,024 evaluated, not a global optimum over the unrestricted parameter space. `max_depth = None` appears in 17 of 20 best configurations (Table 10), which establishes unconstrained tree growth as the default recommendation. The three exceptions, Ionosphere, Credit-G and House Sales, are all best at depth 10, and on none of them is `max_depth` the sole bottleneck: the first two are `max_features`-dominant, and House Sales carries the lowest `max_depth`  $\eta^2$  of any regression dataset except Diabetes. Partial depth constraint thus acts as implicit regularization where the optimal decision boundary has limited complexity. `min_samples_leaf = 1` is optimal in 16 of 20 datasets, so minimal leaf regularization is preferable when depth is unconstrained. `n_estimators` ranges from 50 to 500 across the best configurations, yet its negligible  $\eta^2$  means that variation carries little practical penalty: 100 to 200 trees suffice for most deployments.

For classification, best weighted F1 ranges from 0.7563 on Vehicle to 0.9942 on Banknote, median 0.9605; the three lowest scores, on Vehicle, Covertype and Credit-G, all come from datasets with many classes, class imbalance or mixed-type features. For regression, best  $R^2$  spans 0.4456 on Diabetes to 0.9466 on Bike Sharing, median 0.8267.

## 5 Discussion

The results establish a consistent sensitivity hierarchy across 20 benchmark datasets, with `max_depth` and `max_features` accounting for the dominant share of performance variance in both task types.

### 5.1 Theoretical Interpretation of Sensitivity Rankings

The bias-variance-correlation decomposition of Eq. (3) accounts for the observed ranking. The five hyperparameters do not act on the same number of terms: as the braces in Section 2.1 show, `max_depth` alone governs two additive channels, while `max_features` and `n_estimators` each reach one. That asymmetry predicts a hierarchy with `max_depth` at the top, and the  $\eta^2$  estimates match it: the dominance gap over the second-ranked parameter is 0.12  $\eta^2$  units in classification and 0.61 in regression.

For `n_estimators`, the  $1/T$  factor in Eq. (3) means that increasing from 50 to 500 trees reduces the uncorrelated variance term by 90%, yet the correlated floor  $\rho_h \sigma_h^2$  is entirely unaffected by ensemble size. Once  $T \geq 100$ , the uncorrelated term is already small relative to the correlated floor, making further increases negligible; the mean  $\eta^2 < 0.005$  measures that ceiling. For `min_samples_split`, the redundancy with `min_samples_leaf` follows from the split feasibility constraint: a node can be split only if both resulting children retain at least `min_samples_leaf` observations, so a parent requires at least  $2 \times \text{min\_samples\_leaf}$  samples irrespective of

**Table 10:** Best hyperparameter configurations per dataset (ranked by primary metric on 10-fold CV), selected from the 1,024 configurations of the tested grid rather than from the unrestricted parameter space.

| Dataset       | Task | n_est | max_d | max_f | mss | msl | Metric | Score  |
|---------------|------|-------|-------|-------|-----|-----|--------|--------|
| Iris          | Cls  | 100   | None  | 1.0   | 5   | 1   | F1     | 0.9597 |
| Wine          | Cls  | 50    | None  | sqrt  | 2   | 1   | F1     | 0.9889 |
| Ionosphere    | Cls  | 50    | 10    | sqrt  | 2   | 1   | F1     | 0.9451 |
| Vehicle       | Cls  | 100   | None  | sqrt  | 5   | 2   | F1     | 0.7563 |
| Breast Cancer | Cls  | 50    | None  | 0.5   | 5   | 1   | F1     | 0.9612 |
| Credit-G      | Cls  | 500   | 10    | 0.5   | 10  | 1   | F1     | 0.7659 |
| Banknote      | Cls  | 50    | None  | sqrt  | 2   | 2   | F1     | 0.9942 |
| Digits        | Cls  | 200   | None  | log2  | 2   | 1   | F1     | 0.9811 |
| Segment       | Cls  | 500   | None  | 0.5   | 2   | 1   | F1     | 0.9809 |
| Phoneme       | Cls  | 500   | None  | sqrt  | 2   | 1   | F1     | 0.9137 |
| Covertypes    | Cls  | 500   | None  | 0.5   | 2   | 1   | F1     | 0.7610 |
| Pol           | Cls  | 100   | None  | 0.5   | 2   | 1   | F1     | 0.9869 |
| Diabetes      | Reg  | 50    | None  | 0.5   | 2   | 10  | $R^2$  | 0.4456 |
| Abalone       | Reg  | 500   | None  | 0.5   | 20  | 5   | $R^2$  | 0.5635 |
| Kin8nm        | Reg  | 500   | None  | 1.0   | 2   | 1   | $R^2$  | 0.7264 |
| Elevators     | Reg  | 500   | None  | 1.0   | 2   | 1   | $R^2$  | 0.8416 |
| Bike Sharing  | Reg  | 500   | None  | 0.5   | 2   | 1   | $R^2$  | 0.9466 |
| Houses        | Reg  | 500   | None  | 0.5   | 2   | 1   | $R^2$  | 0.8268 |
| Calif. Hous.  | Reg  | 500   | None  | log2  | 2   | 1   | $R^2$  | 0.8266 |
| House Sales   | Reg  | 50    | 10    | 0.5   | 5   | 1   | $R^2$  | 0.8701 |

n\_est: n\_estimators; max\_d: max\_depth; max\_f: max\_features; mss: min\_samples\_split; msl: min\_samples\_leaf. Cls: classification; Reg: regression. Score = weighted F1 (Cls) or  $R^2$  (Reg).

`min_samples_split`. Any setting with `min_samples_split`  $\leq 2 \times \text{min\_samples\_leaf}$  is therefore inert, because the leaf constraint binds first. In the grid used here this covers 9 of the 16 `min_samples_split`–`min_samples_leaf` level combinations, leaving only 7 in which `min_samples_split` can alter tree growth at all. This explains its near-zero  $\eta^2$  structurally, without reference to the datasets studied. Averaged over datasets the two parameters together account for 2.8% of total performance variance in classification and 1.3% in regression, though the classification figure conceals real spread: on Breast Cancer, where `min_samples_split` reaches  $\eta^2 = 0.101$ , the pair accounts for 12.2%.

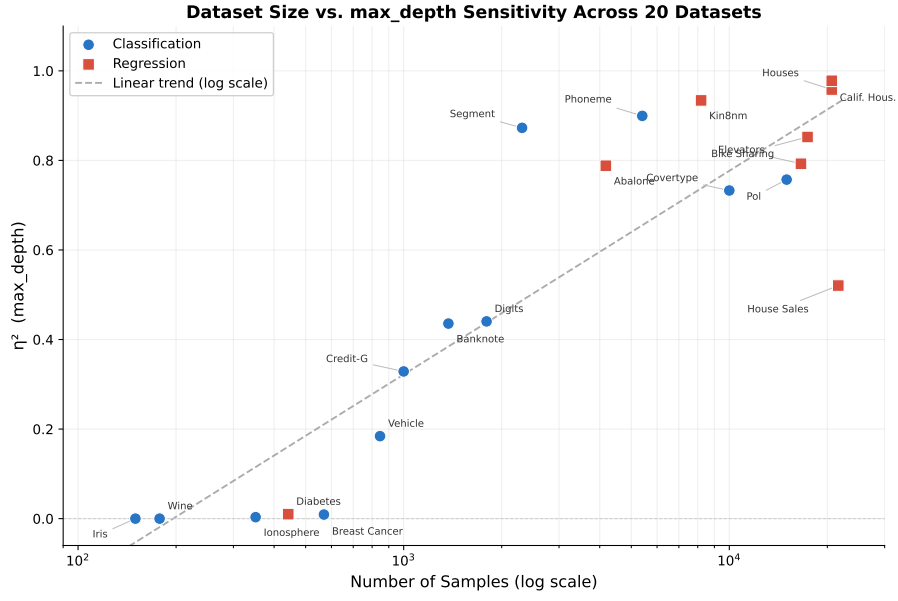
The classification–regression divergence is consistent with target function geometry, on an interpretation this design does not test directly. A depth-5 tree offers at most 32 distinct constant predictions (Section 2.1), a severe constraint on a continuous response surface but not necessarily on a class label: a classification tree may still assign the majority of instances correctly across a complex boundary, because class boundaries are often lower-dimensional than the full response surface. Depth constraints should therefore cost more in regression, as the 87% gap in mean  $\eta^2$  shows. The inverse pattern in `max_features` follows: once depth is unconstrained in regression, deep trees compensate for restricted feature sets at any given split, so `max_features` contributes less marginal gain.

This finding diverges from the conclusion of van Rijn and Hutter [14], who identified `max_features` and `min_samples_leaf` as the primary RF hyperparameters using functional ANOVA on OpenML meta-data. Two methodological differences account for this divergence. First, their surrogate model was fitted on performance records in which `max_depth = None` is the scikit-learn default: when most training configurations already use unconstrained growth, the surrogate observes little variation in depth, systematically underestimating its sensitivity. Second, their benchmark comprised exclusively classification tasks from OpenML-CC18, omitting the regression settings where `max_depth` dominance is most pronounced. Primary experimental control removes both limitations here. The two sets of numbers remain measurements of different quantities, though: a surrogate importance fitted to observational records and an  $\eta^2$  from a controlled enumeration answer one question under two designs, so the comparison is conceptual rather than head-to-head.

## 5.2 Dataset Moderators of Sensitivity

Two dataset characteristics systematically moderate hyperparameter sensitivity, both visible in the cross-dataset  $\eta^2$  heatmap (Figure 2).

**Sample size and max\_depth sensitivity.** `max_depth`  $\eta^2$  increases with sample size (Figure 7). The mechanism is data-driven tree depth: with more observations, unconstrained trees grow deeper, widening the performance gap between `max_depth = None` and `max_depth = 5`. The trend is monotone within classification ( $\rho = 0.942$ ), from Iris ( $n = 150$ ,  $\eta^2 \approx 0.000$ ) to Phoneme ( $n = 5,404$ ,  $\eta^2 = 0.899$ ), and class count is not the driver: Segment and Phoneme reach comparable sensitivity,  $\eta^2 = 0.873$  and 0.899, despite carrying 7 and 2 classes respectively.



**Figure 7:** Scatter plot of  $\text{max\_depth } \eta^2$  against dataset size (log scale) for all 20 benchmark datasets. Classification (circles, blue) and regression (squares, red) follow a broadly increasing trend (Spearman  $\rho = 0.834$ ,  $p < 0.001$  across all 20 datasets). The dashed line shows a linear trend fitted on the log-scale axis. Iris ( $n = 150$ ,  $\eta^2 \approx 0.000$ ) and Phoneme ( $n = 5,404$ ,  $\eta^2 = 0.899$ ) bracket the classification range; Diabetes ( $n = 442$ ,  $\eta^2 = 0.010$ ) and California Housing ( $n = 20,640$ ,  $\eta^2 = 0.978$ ) bracket the regression range.

In regression the size trend is present but statistically weak ( $\rho = 0.419$ ,  $p = 0.301$ ), for the opposite reason: every regression dataset with  $n \geq 4,000$  already sits above  $\eta^2 = 0.50$ , leaving little room for size to discriminate. Diabetes (442 samples,  $\eta^2 = 0.010$ ) is the only regression dataset below the ceiling, and California Housing (20,640 samples,  $\eta^2 = 0.978$ ) sits at its top. Sample size is therefore an informative predictor in classification and a saturated one in regression.

**Feature dimensionality and max\_features sensitivity.** The highest `max_features`  $\eta^2$  values occur at intermediate dimensionality, between 10 and 34 features: Wine, Ionosphere and Breast Cancer all exceed  $\eta^2 = 0.40$ . Sensitivity falls away at both extremes. Phoneme and Banknote, with 5 and 4 features, drop below  $\eta^2 = 0.09$  because `sqrt` and `1.0` converge to the same subset size. Digits and Covtype, with 64 and 54 features, show only moderate sensitivity at  $\eta^2 = 0.365$  and  $0.144$ , suggesting that information content per feature decreases as dimensionality grows, reducing the performance impact of any particular feature subset strategy.

### 5.3 Practical Tuning Recommendations

The  $\eta^2$  effect sizes from this study convert the qualitative importance rankings of prior work [10, 14] into quantitative proportions, enabling practitioners to allocate tuning effort in proportion to measured variance.

1. **Leave unconstrained: max\_depth.** `max_depth` carries the highest mean  $\eta^2$  in both task types (Tables 4 and 6), but what follows from that is that the default is right, not that the parameter repays a sweep: `max_depth = None` is best on 17 of 20 datasets and `max_depth = 5` is significantly worse on 18 of 20 (Table 9). Leave depth unconstrained and spend no evaluations on it: on the six datasets with  $n > 10,000$ , `None` is both the scikit-learn default and the best-performing level, so a depth sweep there returns nothing. Constraining depth costs most in regression, where the mean  $\eta^2$  reaches 0.729. Diabetes is the one dataset on which the choice is immaterial, the Friedman test not rejecting for `max_depth` ( $p = 0.108$ ), and partial constraint pays only where the optimal boundary has limited complexity: Ionosphere, Credit-G and House Sales are best at depth 10.
2. **Tune first: max\_features.** This is where the tuning gain sits. The `sqrt` default is best on only 5 of 20 datasets against 17 of 20 for the `max_depth` default, so a sweep here changes the selected configuration far more often. Priority is higher for classification and for datasets with more than 10 features. `log2` and `sqrt` are broadly interchangeable, significant on only 9 of 20 datasets, so a two-level sweep of `{sqrt, 1.0}` captures the dominant contrast at two evaluations.
3. **Tune second, conditionally: min\_samples\_leaf.** Relevant mainly on small datasets ( $n < 1,000$ ), where it should be swept alongside `max_features` rather than after it, and wherever depth is left unconstrained, since leaf regularization is then the primary remaining complexity control. Its default of 1 is best on 16 of 20 datasets.



4. **Fix at defaults: `n_estimators` and `min_samples_split`.** Mean  $\eta^2 < 0.025$  across all 20 datasets for both parameters: statistically detectable, practically negligible. Set `n_estimators` to 100–200 and `min_samples_split` to 2; tuning either parameter is unlikely to yield measurable performance gains on any dataset resembling those studied here.

## 5.4 Limitations

The following limitations qualify the results.

**Restricted parameter set.** The design manipulates five of the tuning arguments exposed by the implementation and holds the remainder at their defaults (Section 3.1). Every  $\eta^2$  reported here is therefore a share of the variance generated within that five-parameter space, not within the full configuration space of the estimator. `ccp_alpha`, `max_leaf_nodes` and `max_samples` are expected to act through channels already represented, untested here, while `bootstrap` and `criterion` may carry effects a design admitting them would attribute away from the parameters studied. The ranking holds among the five parameters practitioners most commonly tune; it is not a claim that no other argument matters.

**Single cross-validation replicate.** Each configuration was evaluated once, under a 10-fold split generated from `random.state = 42`, so the design does not separate variance attributable to hyperparameters from variance attributable to that particular partition. Because  $\eta^2$  is computed over 1,024 configurations sharing it, the partition acts as a common factor rather than as independent per-configuration noise, and the large effects for `max_depth` are far too big to be an artefact of it. The near-zero values for `n_estimators` and `min_samples_split` need more caution, and the bootstrap intervals of Tables 4 and 6 do not supply it, since they resample across datasets rather than across partitions: a different partition would shift the very values they resample. Those values support the practical conclusion that tuning the two parameters is not worthwhile; establishing that their true effect is null would require repeated cross-validation over multiple seeds.

**Discrete parameter grid and grid-induced contrast.** The four levels per parameter do not cover the full continuous parameter space; effects at intermediate values (e.g., `max_depth`  $\in \{7, 15\}$ ) are unobserved, and  $\eta^2$  estimates reflect the grid used and may differ under finer resolution. In particular, the inclusion of `max_depth` = 5 creates a large performance gap to the next level (depth 10), which contributes to the high  $\eta^2$  observed for `max_depth`. Had the grid begun at depth 10, the measured effect would be smaller. This is a known property of ANOVA-based sensitivity measures: effect size depends on the range of levels tested [8]. The grid spans the range a practitioner would plausibly consider, and a finer eight-level one would require  $8^5 = 32,768$  configurations per dataset.

**Ordinal encoding.** Categorical features in datasets such as `credit-g` and `abalone` were encoded ordinally, which may introduce a spurious ordinal relationship. Tree-based methods are generally robust to monotone transformations, but target encoding or binary encoding could yield different sensitivity profiles for these

datasets.

**Covertypes subset.** The stratified sample of 10,000 instances from 581,012 preserves class proportions but may not represent the full dataset’s spatial and feature distribution. Sensitivity results for Covertypes should be interpreted as applicable to the subset, not the full dataset.

**scikit-learn implementation.** The results are specific to `RandomForestClassifier` and `RandomForestRegressor` in scikit-learn 1.7.2. Alternative implementations (e.g., R `randomForest`, `ranger`) differ in default splitting rules and parallelization, which could affect the relative sensitivity of parameters such as `max_features`.

## 6 Conclusion

A full factorial design over 1,024 configurations, evaluated on 20 benchmark datasets, expresses RF hyperparameter importance as measured shares of performance variance. Four findings follow.

First, `max_depth` dominates in both task types, and its scikit-learn default is what dominates: `max_depth = None` is the best setting on 17 of 20 datasets and `max_depth = 5` is significantly worse on 18 of 20. Second, each dataset falls into one of three sensitivity regimes named by its dominant parameter, `max_depth`-dominant (13 datasets), `max_features`-dominant (6) and `min_samples_leaf`-dominant (1), and which regime applies is predicted in interpretable directions by sample size (Spearman  $\rho = 0.834$  against  $\eta^2(\text{max\_depth})$ ,  $p < 0.001$ ) and by feature dimensionality. Third, the profiles diverge by task type: `max_depth` dominance is 87% larger in regression than in classification, while `max_features` matters proportionally more in classification ( $\eta^2 = 0.268$  against 0.119). Fourth, the two leading parameters carry opposite practical consequences. The `max_depth` default should be left in place, whereas the `max_features` default is best on only 5 of 20 and is therefore the parameter that repays a sweep; `n_estimators` and `min_samples_split`, at mean  $\eta^2 < 0.025$  throughout, can be fixed at their defaults. All of this holds within the grid enumerated here, not over the unrestricted parameter space.

The same design applied to gradient boosting would answer the corresponding question there. XGBoost, LightGBM and CatBoost share tree depth, feature subsampling and leaf regularization with RF, but add a learning rate and boosting-specific parameters whose sensitivity profiles, and whose interactions with dataset characteristics, remain poorly characterized by primary experimental evidence.

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**Conflict of interest.** The authors declare no conflict of interest.

**Data and code availability.** All benchmark datasets are publicly available through scikit-learn and OpenML [15]; the individual OpenML identifiers are listed in Tables 2 and 3. The experiment script and the complete per-dataset result files, statistical summaries and post-hoc test outputs are available under the MIT licence at <https://github.com/pirasi-kku/rf-hyperparameter-sensitivity>.

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